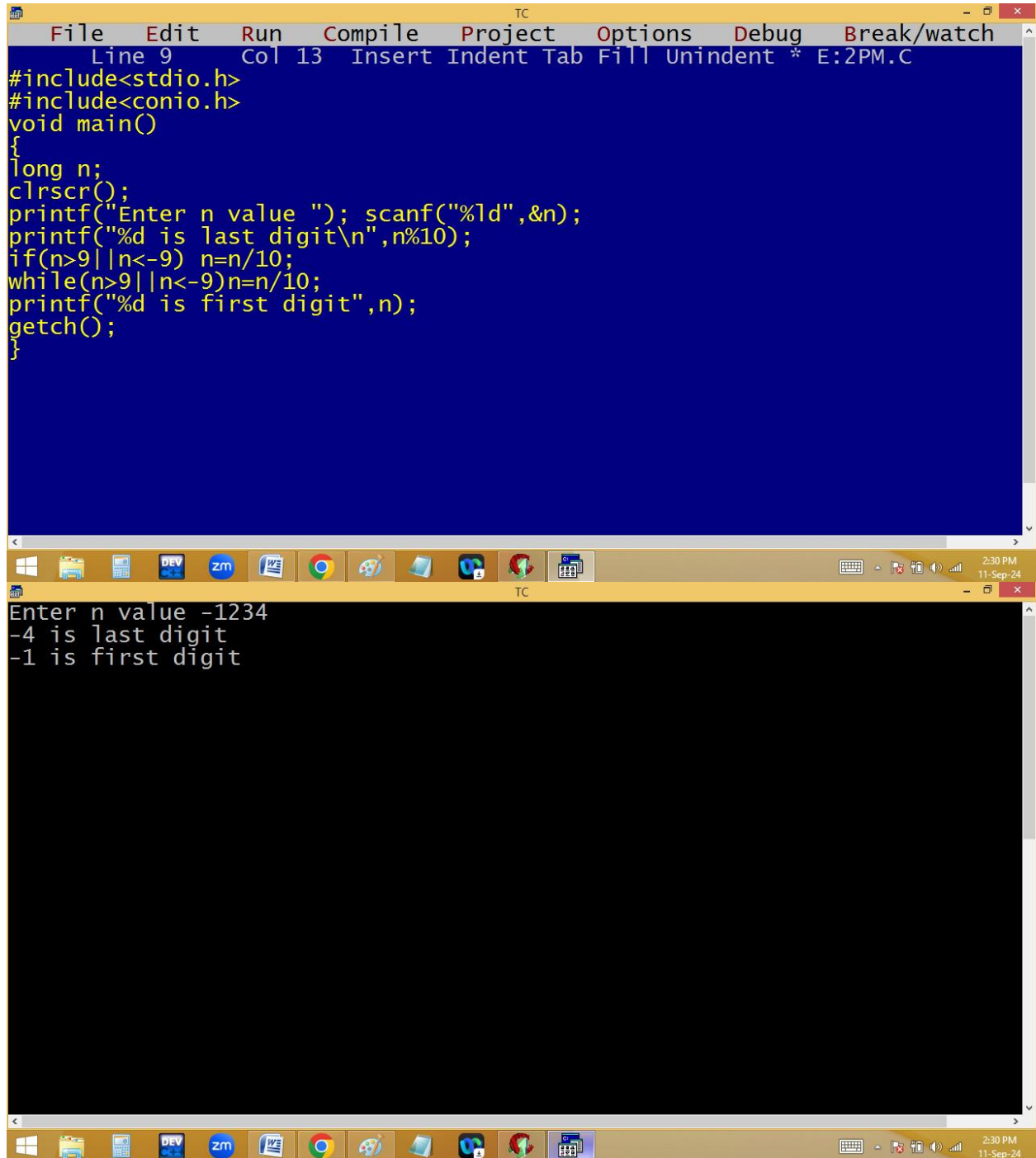


Finding first and last digits given no:

Eg: 2016 → 6 is last digit, 2 is first digit.



The screenshot displays the Turbo C++ (TC) IDE interface. The top window shows the source code for a C program. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, it declares a `long` variable `n`, clears the screen with `clrscr()`, and prompts the user to enter a value for `n` using `scanf("%ld",&n);`. It then prints the last digit using `printf("%d is last digit\n",n%10);`. To find the first digit, it uses a loop: `if(n>9||n<-9) n=n/10;` followed by `while(n>9||n<-9)n=n/10;`, and finally prints the first digit with `printf("%d is first digit",n);`. The program ends with `getch();`. The bottom window shows the program's execution output for the input `-1234`, displaying `-4 is last digit` and `-1 is first digit`. The Windows taskbar at the bottom shows the time as 2:30 PM on 11-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
printf("%d is last digit\n",n%10);
if(n>9||n<-9) n=n/10;
while(n>9||n<-9)n=n/10;
printf("%d is first digit",n);
getch();
}
```

Enter n value -1234
-4 is last digit
-1 is first digit

```
TC
Enter n value 1234
4 is last digit
1 is first digit
```



```
TC
Enter n value 3
3 is last digit
3 is first digit_
```

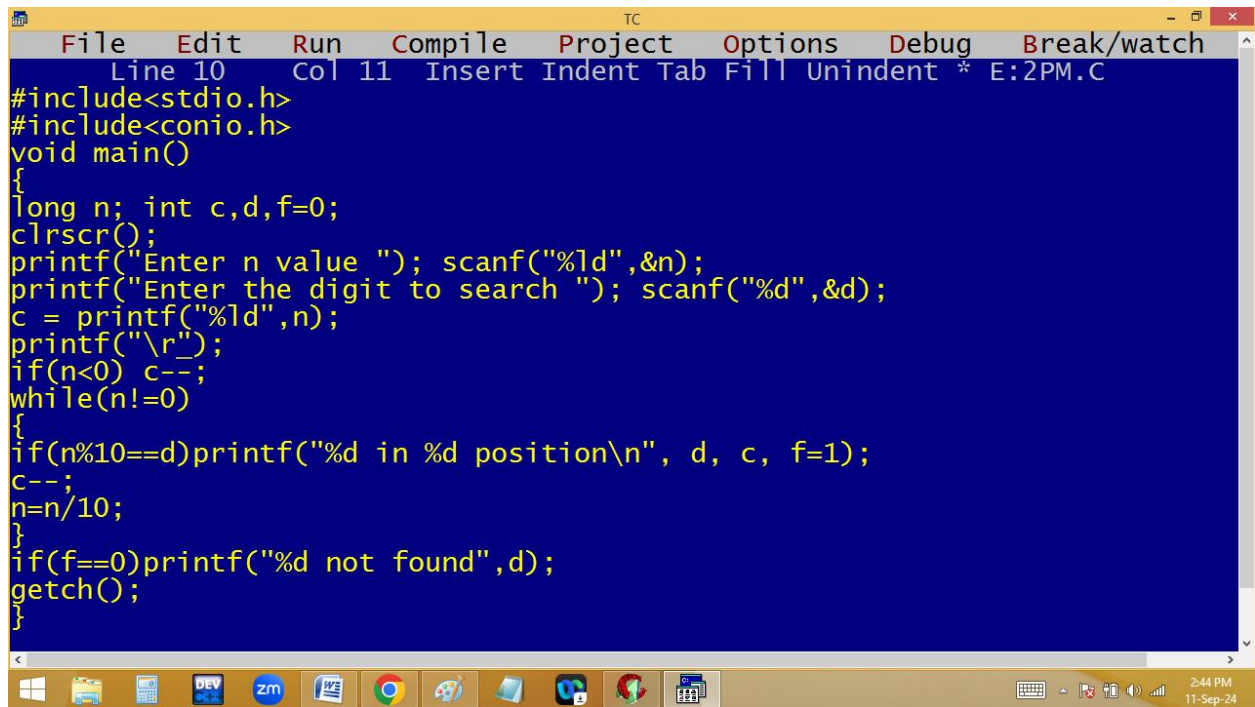
7
printf("%d is last digit\n", n%10);
n=n/10; ✓
while(n>9) n=n/10;
p("%d first digit", n);

17
1927 %10 = 7 is last digit
177

Finding the given digit position in given no.

2017 → 0 position is 2

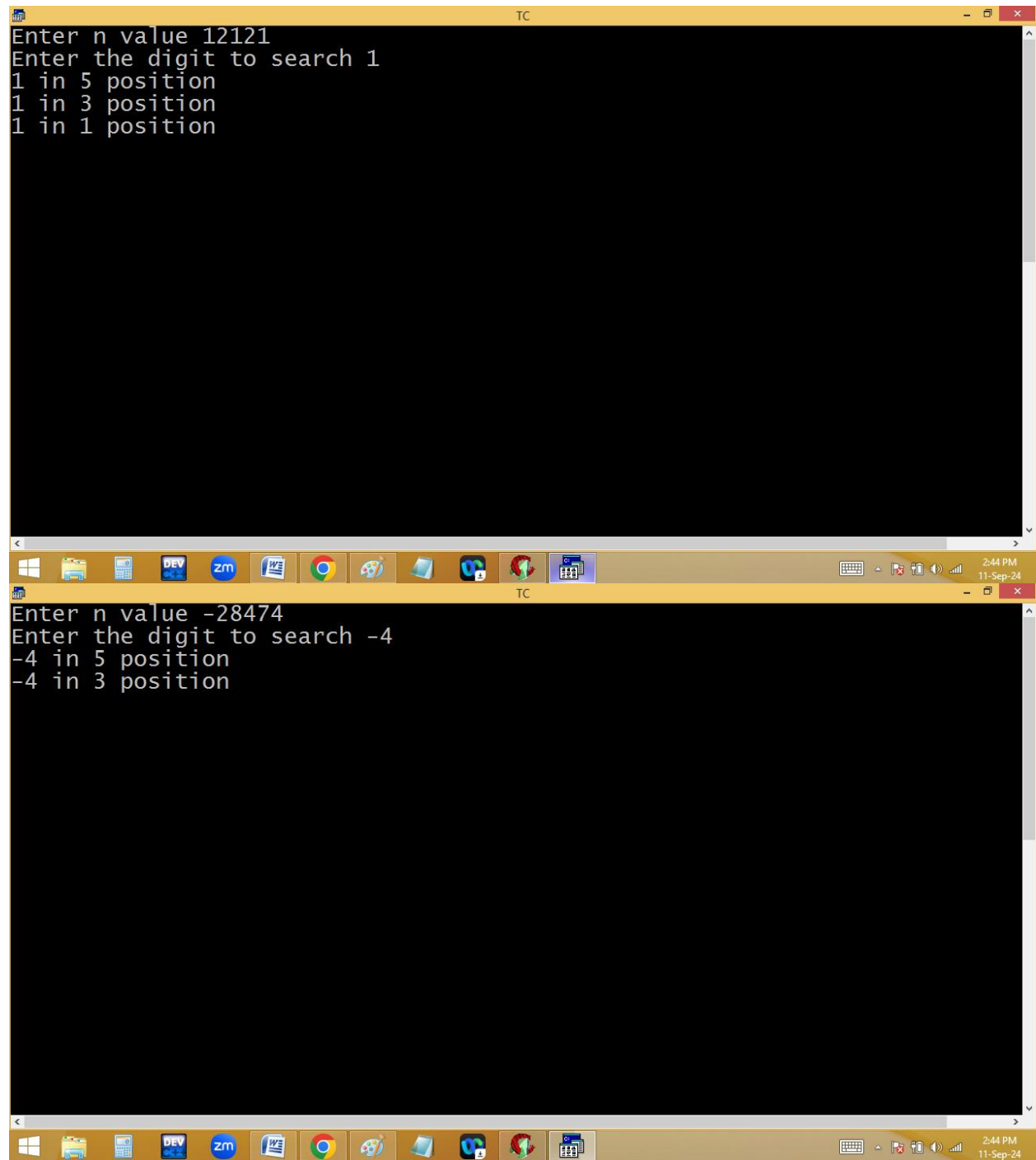
2017 is a 4 digit no



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the bottom of the window displays "Line 10", "Col 11", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int c,d,f=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
printf("Enter the digit to search "); scanf("%d",&d);
c = printf("%ld",n);
printf("\r");
if(n<0) c--;
while(n!=0)
{
if(n%10==d)printf("%d in %d position\n", d, c, f=1);
c--;
n=n/10;
}
if(f==0)printf("%d not found",d);
getch();
}
```

The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock indicating 2:44 PM on 11-Sep-24.



```
TC
Enter n value 1234
Enter the digit to search 9
9 not found
```

`c = p("%d",n);`

`while(n!=0)`

`{`

`if(n%10==d) p(c);`

`c--; n=n/10;`

`}`

n d

$$2714 \% 10 = 4 == 7$$

$$271 \% 10 = 1 == 7$$

$$27 \% 10 = 7 == 7$$

c

4

3

2

1

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a C program named E:2PM.C. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 25 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long m,n; int c=0,d,f=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
printf("Enter the digit to search "); scanf("%d",&d);m=n;
while(m!=0){c++;m=m/10;}
while(n!=0)
{
if(n%10==d)printf("%d in %d position\n", d, c, f=1);
c--;
n=n/10;
}
if(f==0)printf("%d not found",d);
getch();
}
```

The bottom window shows the program's execution output:

```
Enter n value 929579
Enter the digit to search 9
9 in 6 position
9 in 3 position
9 in 1 position
```

The IDE's taskbar at the bottom shows various application icons, including Windows Explorer, DEV C++, Zoom, Word, Chrome, and others. The system clock in the bottom right corner indicates the time is 2:47 PM on 11-Sep-24.

```

Enter n value 34535
Enter the digit to search 0
0 not found_

```

```

while(m!=0){c++;m/=10};
while(n!=0)
{
if(n%10==d) p(c);
c--; n=n/10;
}

```

$\frac{n}{10}$	=	$\frac{n}{10}$	=	$\frac{d}{10}$	$\frac{c}{10}$
34535		2714		5	1
		271		4	2
		27		1	3
		2		7	4

Finding max, min digits in given no.

Eg: 40372 → 0 min, 7 max

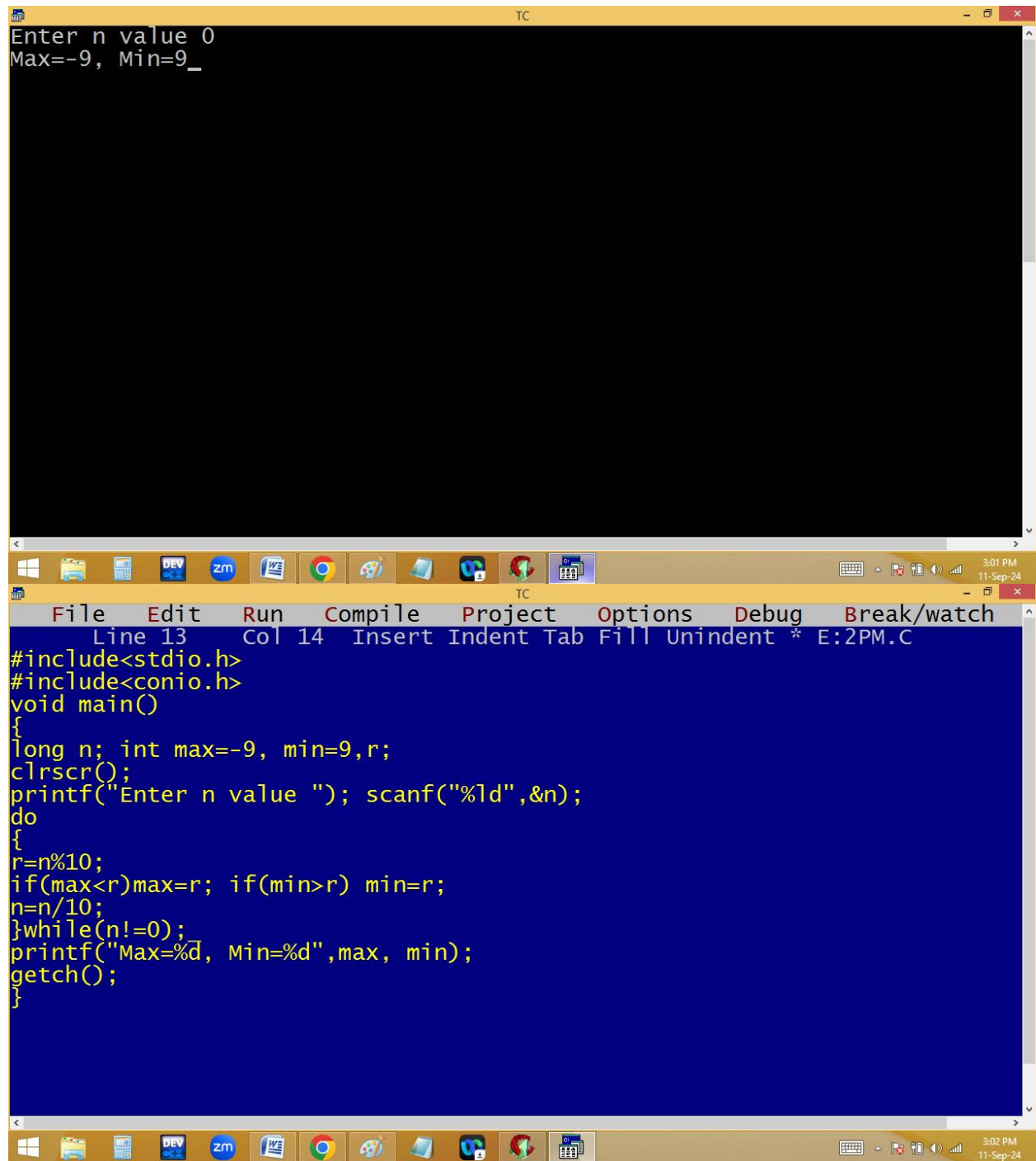
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a C program named 'E:2PM.C'. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int max=-9, min=9,r;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
while(n!=0)
{
r=n%10;
if(max<r)max=r; if(min>r) min=r;
n=n/10;
}
printf("Max=%d, Min=%d",max, min);
getch();
}
```

The bottom window shows the program's execution. It prompts the user to 'Enter n value' and the input '203914' is shown. The output of the program is 'Max=9, Min=0'. The Windows taskbar at the bottom indicates the time is 3:00 PM on 11-Sep-24.

```
TC
Enter n value 1234
Max=4, Min=1_
```

```
TC
Enter n value -3826
Max=-2, Min=-8
```



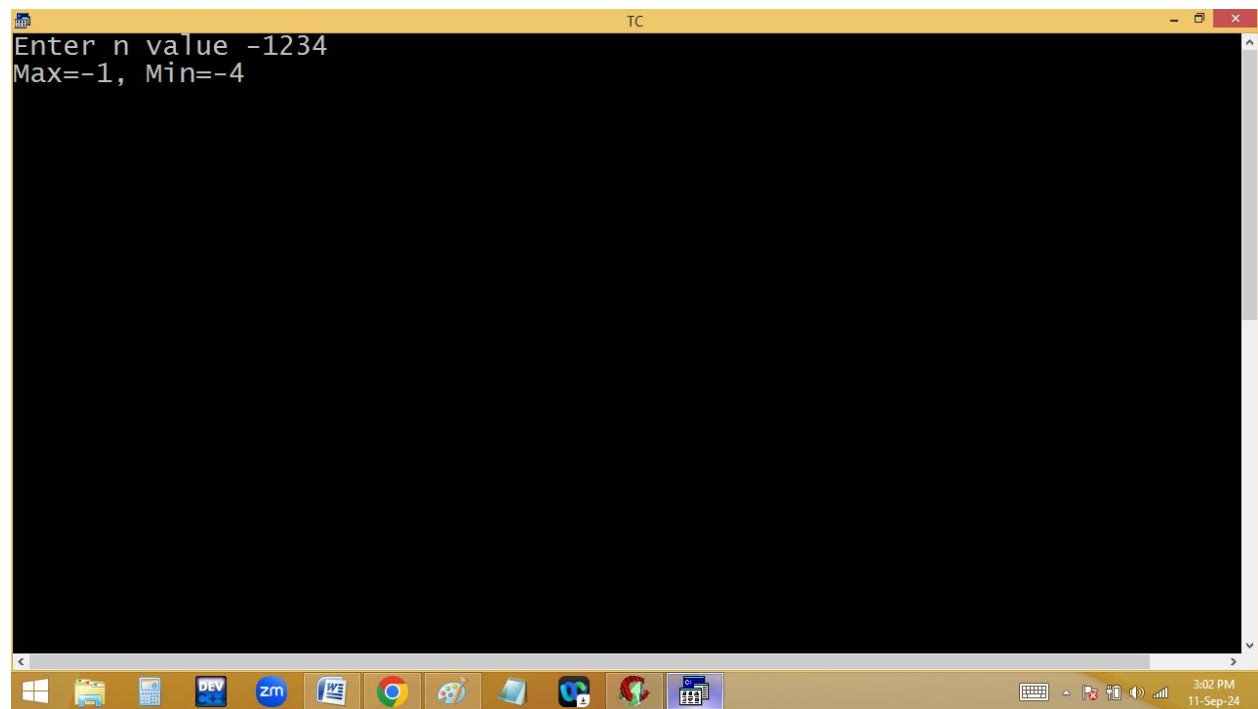
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the output of the program, which prompts the user to "Enter n value 0" and shows the initial values "Max=-9, Min=9_". The bottom window shows the source code for the program, which is a C program that calculates the maximum and minimum values of a number n by repeatedly dividing it by 10. The code includes standard headers, a main function, and a loop that continues until n becomes 0. The program uses scanf to take input and printf to display the results.

```
Enter n value 0
Max=-9, Min=9_

#include<stdio.h>
#include<conio.h>
void main()
{
    long n; int max=-9, min=9,r;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    do
    {
        r=n%10;
        if(max<r)max=r; if(min>r) min=r;
        n=n/10;
    }while(n!=0);
    printf("Max=%d, Min=%d",max, min);
    getch();
}
```

```
TC
Enter n value 0
Max=0, Min=0
```

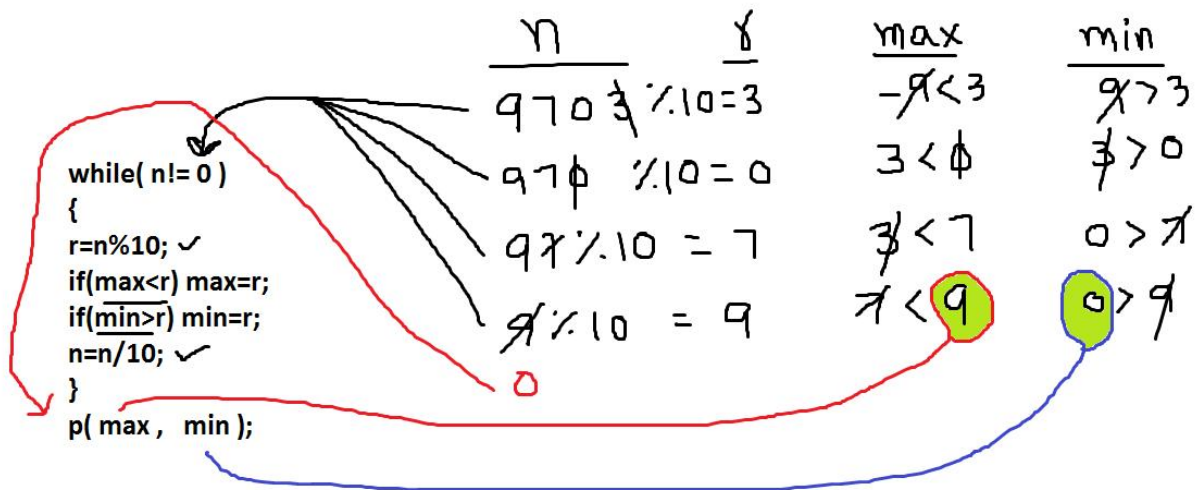
```
TC
Enter n value 2345
Max=5, Min=2_
```



```

Enter n value 8
Max=8, Min=8

```



Finding no of even/odd/zero digits in given no.

Eg: 1023 → 1 even, 2 odd, 1 zero

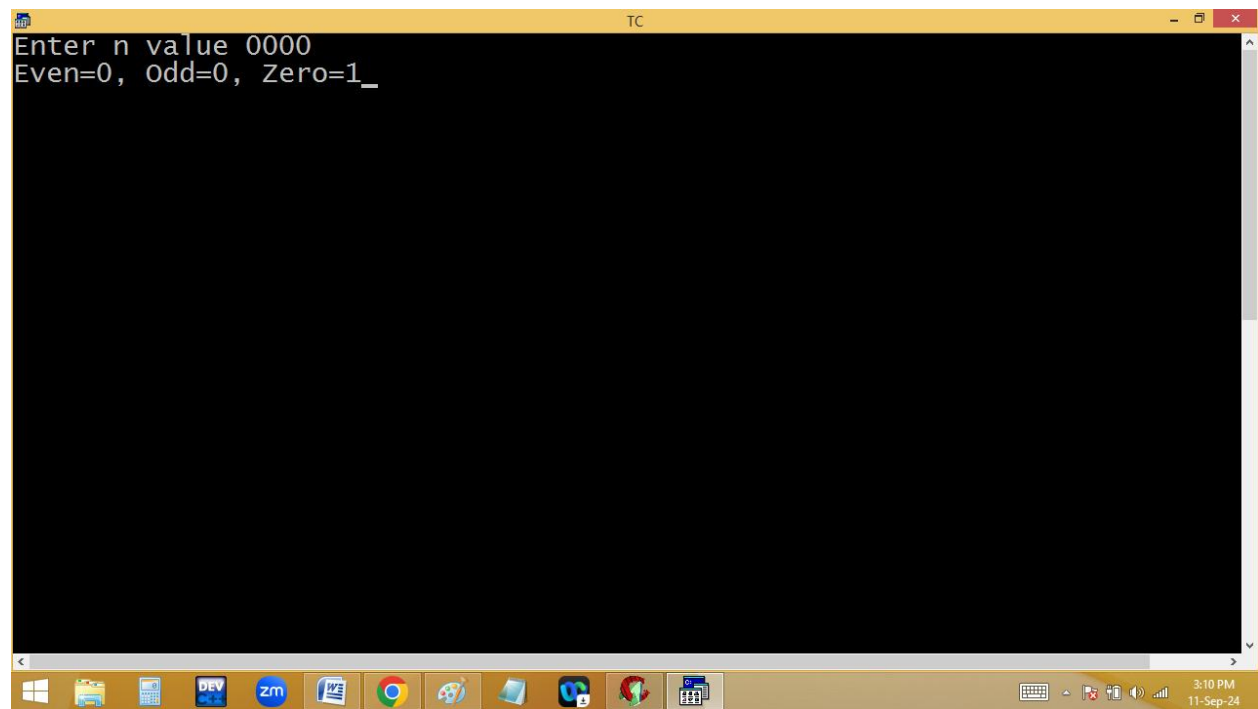
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int e,o,z,r;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
e=o=z=0;
do
{
r=n%10;
if(r==0)z++; else if(r%2==0)e++;else o++;
n=n/10;
}while(n!=0);
printf("Even=%d, Odd=%d, Zero=%d",e,o,z);
getch();
}

Enter n value 1023
Even=1, Odd=2, Zero=1

TC
3:09 PM
11-Sep-24
```

```
TC
Enter n value 2468
Even=4, Odd=0, Zero=0
```

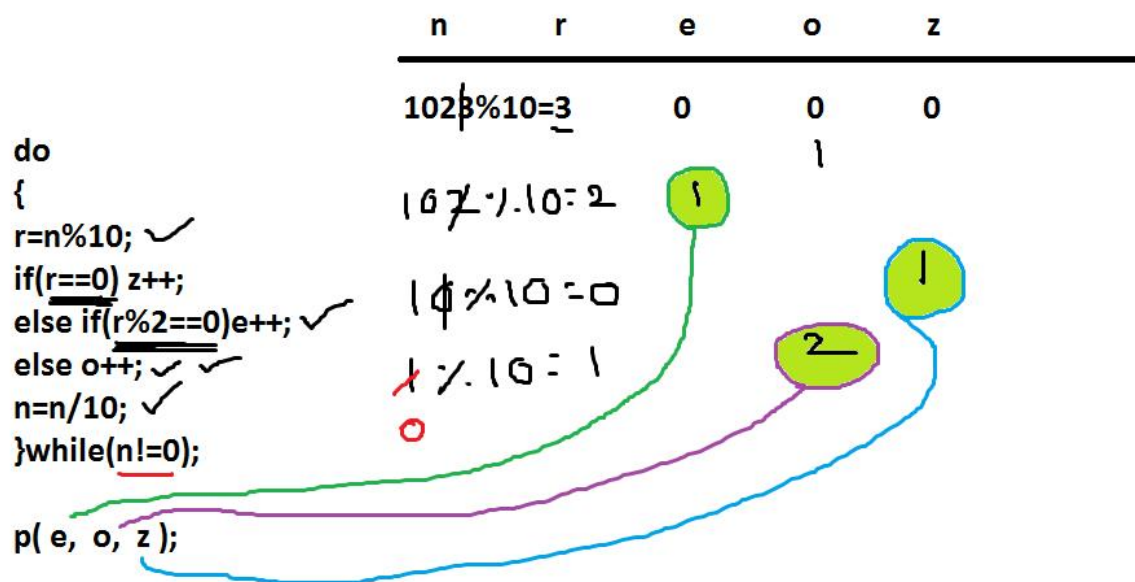
```
TC
Enter n value 1357
Even=0, Odd=4, Zero=0_
```

```

Enter n value -1023
Even=1, Odd=2, Zero=1_

```



Reverse no:

n

257

$7 \times 10 = 70$

$+ \underline{5}$

$75 \times 10 = 750$

$+ \underline{2}$

752

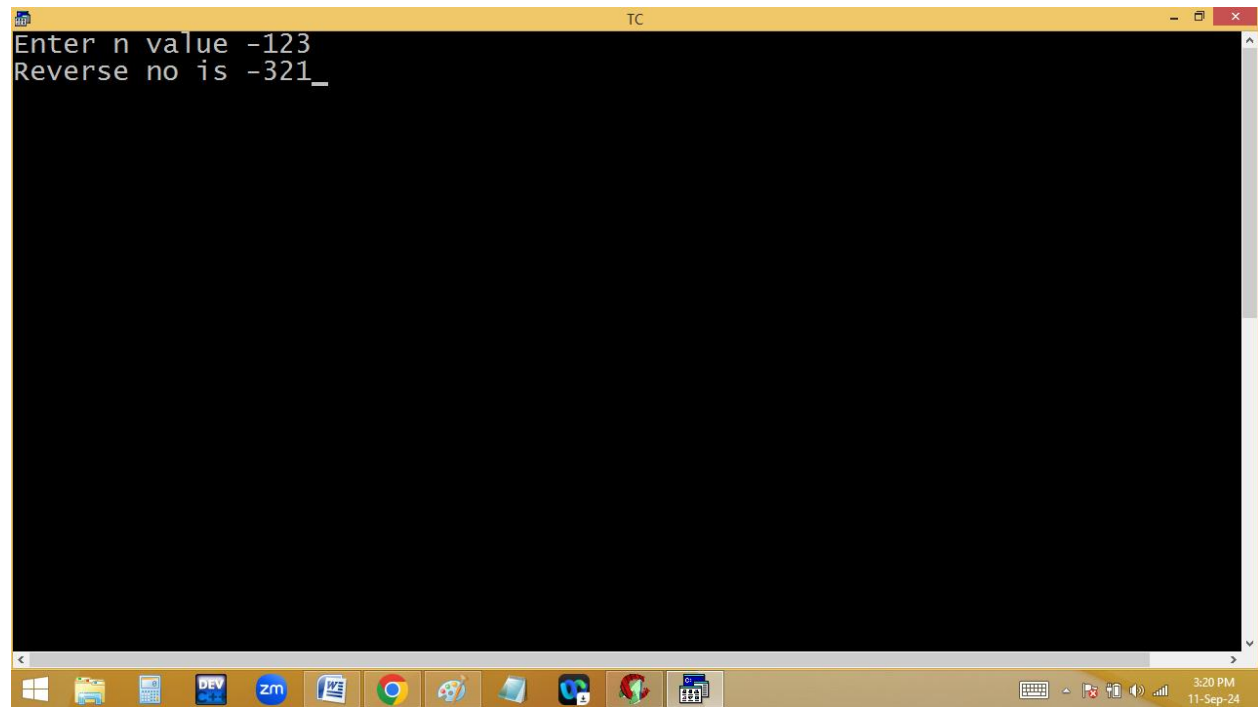
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a program that reverses a number. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 33 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n,rev=0; int r;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
while(n!=0)
{
r=n%10; rev=rev*10+r; n=n/10;
}
printf("Reverse no is %ld",rev);
getch();
}
```

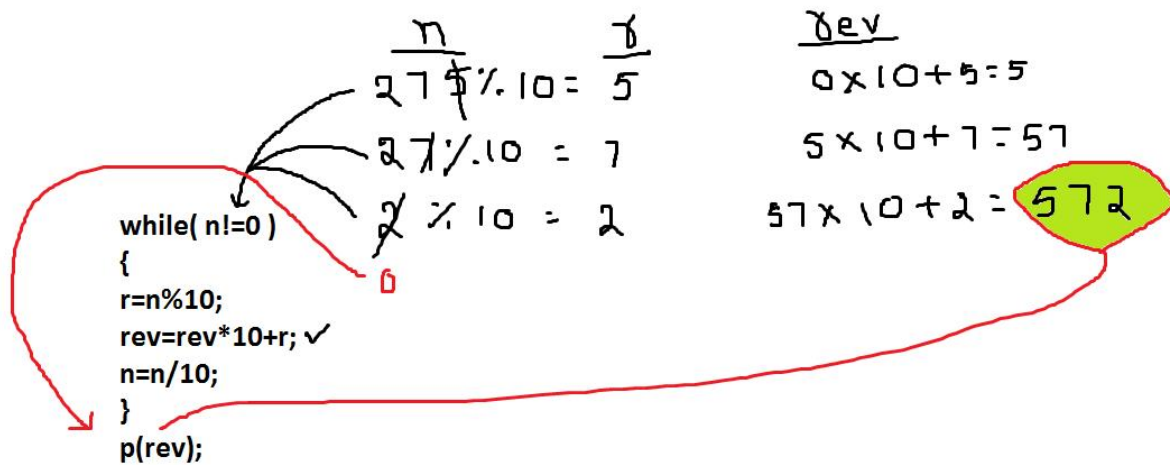
The bottom window shows the execution of the program. It prompts the user to enter a value, and the user has entered 123. The program then outputs the reversed number, 321.

```
Enter n value 123
Reverse no is 321
```

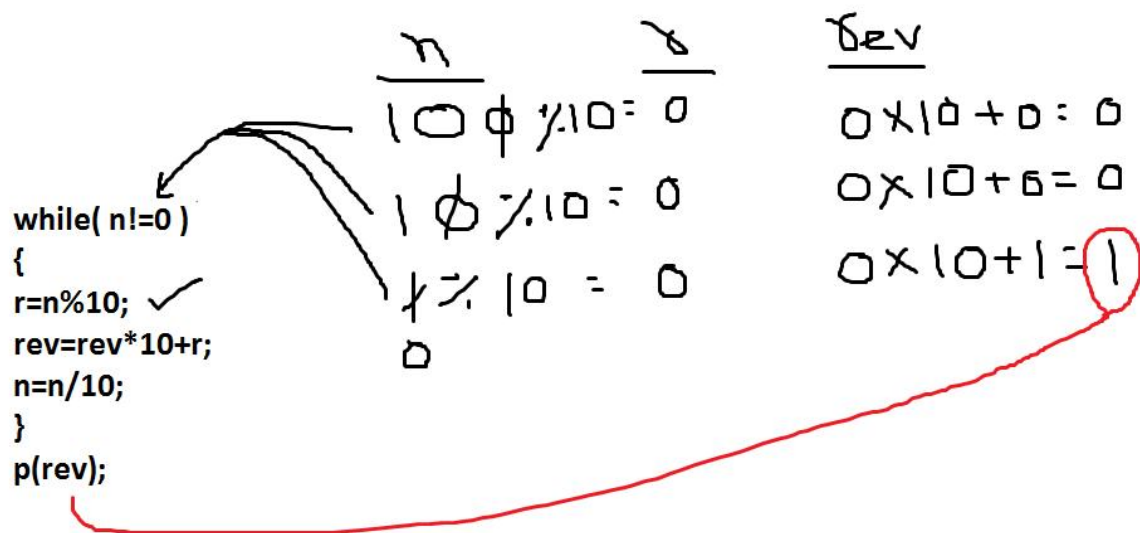
The Windows taskbar at the bottom of the screen shows the time as 3:20 PM on 11-Sep-24. Various application icons are visible in the taskbar, including Windows Explorer, DEV C++, Zoom, and Google Chrome.



```
Enter n value 0
Reverse no is 0
```



```
Enter n value 100
Reverse no is 1
```



Method 2:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 69 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n,rev=0; int r;
clrscr();
printf("Enter n value "); scanf("%ld",&n); if(n<0) printf("-",n=-n);_
do
{
r=n%10; printf("%d",r); n=n/10;
}while(n!=0);
getch();
}

Enter n value 100
001_

TC
3:31 PM
11-Sep-24
```



```
TC
Enter n value -100200
-002001
```

```
TC
Enter n value 123
321_
```

```
TC
Enter n value -123
-321
```

```
TC
Enter n value 0
0_
```

	<u>n</u>	<u>p(x)</u>
	100	0 % 10 = 0
	10	0 % 10 = 0
	1	1 % 10 = 1
	0	


```

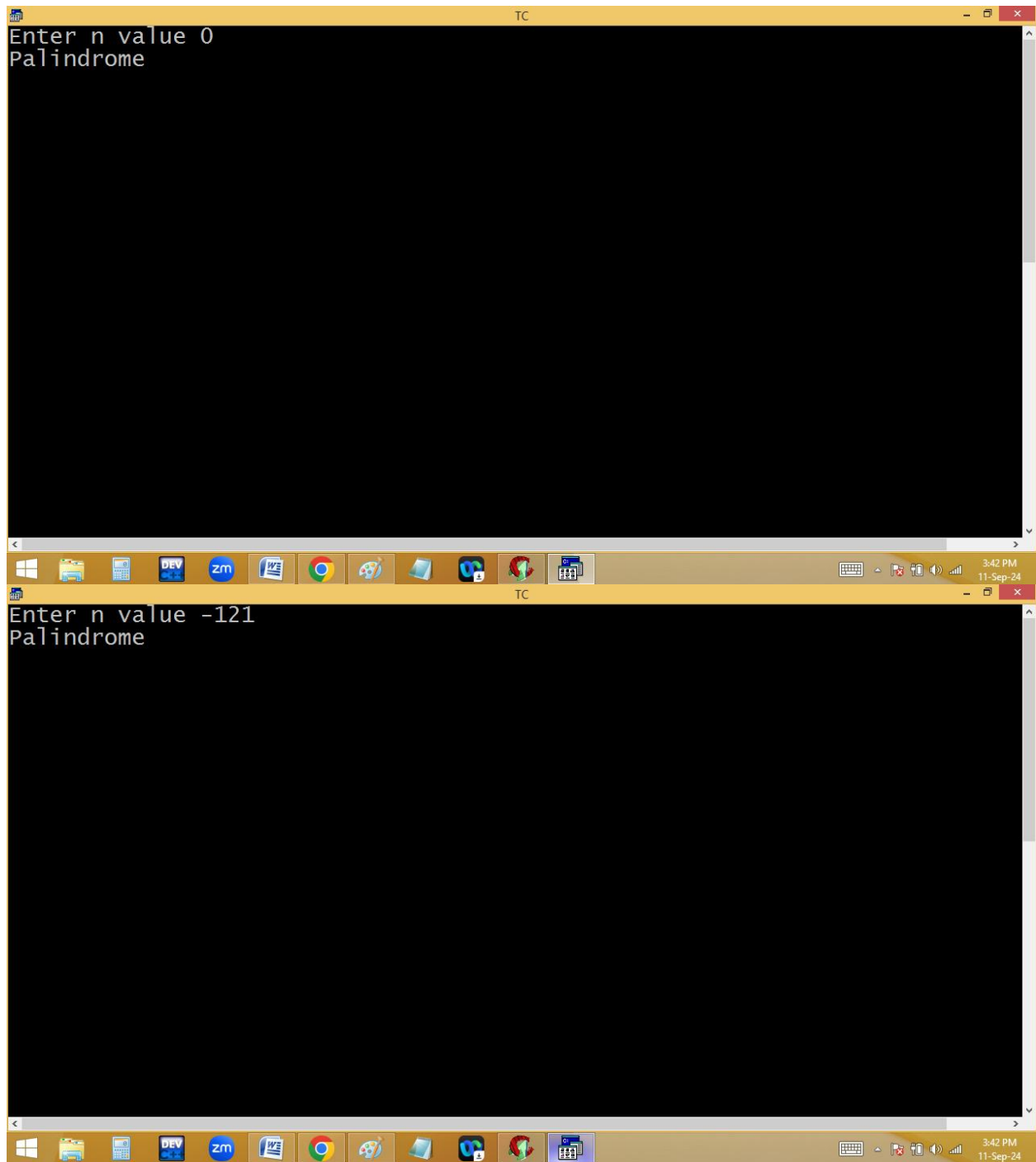
while( n!=0 )
{
  r=n%10; ✓
  rev=rev*10+r;
  n=n/10;
}
p(rev);
  
```

finding palindrome:

121 reverse is 121

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 5 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n,rev=0,m; int r;
clrscr();
printf("Enter n value "); scanf("%ld",&n); m=n;
while(n!=0)
{
r=n%10; rev=rev*10+r; n=n/10;
}
if(m==rev)puts("Palindrome");else puts("Not a Palindrome");
getch();
}

TC
Enter n value 121
Palindrome
```



Home work:

Eg 1: 102 → One Zero Two

Finding Armstrong no:

1 is single digit no $\rightarrow 1^1 = 1$

9 is single digit no $\rightarrow 9^1 = 9$

153 is 3 digit no $\rightarrow 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

1634 is 4 digit no $\rightarrow 1^4 + 6^4 + 3^4 + 4^4 = 1634$

370, 371, 407, 8208,.....

